ITQAN - MICRO-ASSESSMENT AI AGENT

Policy-aware formative assessment and staged educational incident simulation

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GitHub: github.com/alzoubitoqa/micro-assessment-agent | Domain: Education

1. Introduction and problem

Traditional high-stakes examinations can reveal comprehension gaps too late for effective intervention. CodeMentor AI transforms uploaded lecture PDFs into source-grounded micro-assessments, provides contextual hints and simpler retries, and shows instructors topic-level performance signals. In the ITU UAE extension, the same system becomes a laboratory for investigating AI-generated assessment errors, privacy questions and academic integrity false positives.

2. Solution and distinctive contribution

• Assessment: local PDF text extraction, semantic retrieval, Groq LLM question generation and source-chunk validation.

• Adaptation: LangGraph state transitions route wrong answers to hints and validated simplified questions.

• Instructional analytics: SQLite records attempts, feedback and response times; dashboards show final mastery separately from first-attempt performance.

• Governance: a UAE/global policy catalog, three institutional roles, staged incident events and transparent human-review recommendations.

The tool supports Sustainable Development Goal 4 by enabling early feedback while highlighting how human oversight, fair treatment, accountability and privacy matter in education.

Status: working prototype and reproducible offline simulation; not a certified or production-grade institutional system.

Prototype evidence available for review

Artifact	Independent verification possible
Source code	Public GitHub repository with separated lecture RAG, policy retrieval, assessment and governance modules.
Functional tests	Ten automated regression tests for stage order, role visibility, source provenance, question validation, lecture filtering and analytics.
Runnable exercise	Two fictional four-event incident scenarios, with policy references and downloadable audit logs.

3. Technical architecture and AI pipeline

Layer	Implemented pipeline / implementation
SRC / ingestion	Instructor lecture PDF, student answer events; pypdf extraction and page-aware chunks.
Preprocessing	Recursive split (default 400 characters, 60 overlap), local Sentence-Transformers embeddings.
Model / retrieval	ChromaDB lecture collection; top-k scoped by selected lecture; Groq Llama 3.3 70B question generation.
Policies / validation	Source ID verification, semantic Validator Agent, revisited validation after simplification, policy evidence in separate collection.
Orchestration	LangGraph question/hint/simplify workflow; independent incident-role assessments and explicit coordination protocol.
SINK / audit	Streamlit student and instructor portals; SQLite attempt and incident records; downloadable JSON decision traces.

This is a conceptual pipeline mapping to ITU-T Y.3172 components. The recommendation formally concerns machine learning in future telecommunications networks; this educational prototype does not claim Y.3172 certification or network implementation.

4. Multi-agent responsibilities

Agent or role	Specific mandate and interaction
Generator / Validator	Generate source-grounded questions; reject unsupported answers or unverified source IDs.
Adaptive Tutor	Deliver contextual hints, then a simpler question when necessary.
Integrity Officer	Treat fast responses as weak advisory indicators, not proof of misconduct.
Data Protection Officer	Inspect data flow, processing basis, access and retention questions.
Academic Quality Reviewer	Assess question validity, learner impact and correction of faulty records.
Coordinator	Combine independent role assessments after each event; defer final sanction or grade action to a human.

Institutional agents see only events addressed to their roles. The optional policy LLM is constrained to retrieved source IDs, while a deterministic fallback enables offline reproduction.

5. Country-specific knowledge base and policy mapping

The shareable seed catalog contains ten provenance records: UAE legal and strategy references plus UNESCO and ITU guidance. Its initial text consists of labelled curated summaries and source metadata; official UAE PDF page indexing is supported after local download. These summaries must not be misrepresented as the complete national policy corpus.

Source	Relevant issue / legal status
UAE Personal Data Decree-Law 45/2021	Articles 1, 2 and 4: personal-data definitions, scope and processing basis. Applicability varies by institution; statutory law where applicable.
UAE National AI Strategy 2031	AI integration, governance and infrastructure priorities. National strategy; not a statutory compliance certificate.
UNESCO AI Ethics Recommendation (2021)	Paragraphs 43, 53 and 104: audit, impact assessment and strict safeguards around monitoring learners. International, non-binding guidance.
ITU AI Readiness	Data governance, AI policy, infrastructure and innovation metrics. Assessment framework, not binding legislation.

Each catalog item retains an official URL, authority, jurisdiction, type, citation reference and a binding-level label. Policies are stored separately from course lectures to avoid accidental cross-domain retrieval. A vector index can include original PDF page chunks.

6. ITU AI Readiness perspective

Readiness dimension	Implemented evidence / limitation
Data and governance	Local embeddings, scoped retrieval, policy provenance; student retention and access controls remain incomplete.
GenAI and integration	RAG question generation, validation, optional policy assistant and multi-role simulation.
Human oversight and inclusion	No automatic punishment; human decision pending; learner accessibility and fairness studies required.
Infrastructure and sustainability	Local embedding and hosted inference; latency, energy and deployment cost require measurement.

7. Interactive progressive error scenarios

Scenario A - False Academic Misconduct Flag

Stage	New information / expected institutional response
Event 1	Correct first-attempt answer in 2.4 seconds triggers a heuristic flag. Integrity and quality reviewers advise against automatic accusations.
Event 2	Prior legitimate fast responses are disclosed to two roles. They reassess the false-positive risk.
Event 3	Student seeks explanation and correction of stored behavioural data. Privacy officer receives the event and inspects data-handling obligations.
Event 4	Instructor finds no corroborating evidence. All roles contribute to a documented, human-reviewed resolution.

Scenario B - Hallucinated Question and Assessment Failure

Stage	New information / expected institutional response
Event 1	Malformed PDF causes an unsupported generated answer. Academic Quality Reviewer recommends blocking the question.
Event 2	Affected student attempts appear. Quality and Integrity Officers recommend correction without penalizing students.
Event 3	External LLM processing of retrieved lecture text is revealed to the Data Protection Officer.
Event 4	Academic committee requests traceability, corrected analytics and revised validation controls.

8. Transparent decision traces

A SQLite audit stores run ID, event number and timestamp, facts released to each role, role mandate, source IDs and official URLs, succinct public rationale, recommendations, detected corpus gaps and inter-agent coordination. A JSON report exposes this record to instructors. The system does not disclose or retain private model chain-of-thought; no punitive action is automated.

The original incident sequence is intentionally hidden from roles until each event is released. This allows reproducible tests of incremental information, uneven institutional knowledge, and reviewable coordination.

Scenario A - information distribution

Institutional role	Events received as the simulation unfolds
Academic Integrity Officer	Events 1, 2, 3 and 4
Academic Quality Reviewer	Events 1, 2 and 4; does not initially receive the student's data-access inquiry.
Data Protection Officer	Events 3 and 4; receives no initial cheating allegation before disclosure.

Every log entry includes evidence identifiers and an explicit rule or public rationale. Incomplete policy coverage is identified as an uncertainty requiring human verification.

9. Evaluation and reproducibility

Static code review and the pure-Python regression suite were run on 1 November 2026. Ten of ten tests passed: catalog provenance, policy retrieval and classification, scenario availability, event order and absence of future-event leaks, role visibility, audit citations, scoped lecture retrieval and final-mastery analytics. The offline four-stage false-positive scenario completed successfully.

These are functional checks, not a measured question-quality benchmark or evidence of real-world legal compliance. Full Groq/Streamlit deployment and user studies remain unverified.

10. Policy, strategy and standards gaps

• Academic integrity governance: the current catalog does not establish an institution-specific rule on response-time surveillance, escalation or appeals. This is a corpus gap, not proof that no such rule exists.

• Assessment fairness: a source-grounded Validator does not automatically demonstrate cultural, linguistic or disability fairness; independent audits are necessary.

• Data processing: hosted LLM calls can include retrieved lecture excerpts; local embeddings alone do not guarantee data residency or compliance.

• Institutional coordination: formal mandates, access rights and incident response playbooks require validation with real organizations.

11. Limitations and next steps

The prototype has no production authentication, retention enforcement, deployed distributed ITU Lab knowledge base or measured energy impact. Next steps: integrate official full-text documents, benchmark policy retrieval/citation quality, secure institutional access, perform accessibility tests and connect to the ITU Lab infrastructure when authorized.

Primary sources

[1] ITU AI for Good Lab UAE Hackathon: aiforgood.itu.int/event/ai-for-good-sandbox-hackathon-united-arab-emirates/
[2] ITU-T Y.3172 (2019): itu.int/rec/T-REC-Y.3172/en
[3] ITU AI Ready Framework: itu.int/epublications/publication/ai-ready-analysis-towards-a-standardized-readiness-framework
[4] UAE Federal Decree-Law 45/2021: uaelegislation.gov.ae/en/legislations/1972/download
[5] UAE National AI Strategy 2031: ai.gov.ae/strategy
[6] UNESCO Recommendation on the Ethics of AI (2021): unesco.org/en/legal-affairs/recommendation-ethics-artificial-intelligence